

THE SIGNALRUPTURE STRUCTURAL GRAMMAR PREDICTION

A Frozen Prospective Test of Requirement, Durable Capacity, Compensation, Restoration, and Function

SignalRupture (SR)

Prospective Metatheory Validation Protocol — Frozen August 2026

Abstract

This manuscript freezes the first prospective prediction of the SignalRupture (SR) Structural Grammar following its three-domain empirical operationalization in healthcare, electricity, and households. The paper does not predict societal collapse. It predicts a relational sequence. When effective requirement (Q) persistently exceeds durable capacity (D), systems possessing expandable compensatory capacity should initially preserve observable function (F) by increasing sustainable flexibility or extraordinary compensation (C). During this interval, surface function should become an incomplete indicator of durable structural sufficiency. The system should subsequently branch. If durable restoration closes the requirement-capacity gap, extraordinary compensation should decline while function remains stable or improves. If the gap persists and extraordinary compensation becomes constrained, plateaus, or declines without restoration, pre-specified functional outcomes should deteriorate with a temporal lag. The same directional architecture is predicted to recur across heterogeneous domains despite different physical mechanisms. The protocol freezes five primary predictions: Compensated Continuity, Surface–Substrate Divergence, Compensatory Inversion, Restoration, and Cross-Domain Invariance. It also freezes relative temporal ordering, domain translations, falsifiers, scoring rules, and a prohibition on retrospective redefinition. The purpose is to allow future observations to determine whether SR remains a diagnostic grammar, acquires prospective predictive validity, or fails its metatheoretical claim.

Keywords: SignalRupture; structural grammar; prospective prediction; compensated continuity; compensatory inversion; restoration; cross-domain invariance; metatheory; falsifiability.

1. Prediction Status

This is a prospective prediction paper. Its definitions and directional claims are frozen in August 2026 before the principal future outcomes are known. Future revisions may formulate new hypotheses, but they may not silently alter the prediction scored here. The scientific object is not whether a future crisis occurs. The object is whether a stable relational grammar predicts the ordering of requirement, durable capacity, compensation, restoration, and observable function across heterogeneous systems.

Field identity ≠ operational invariance ≠ prospective predictive validation

The three-domain empirical synthesis established operational translation. This paper freezes the next burden of proof: temporal prediction.

2. Canonical Structural Grammar

Symbol	Canonical meaning	Rule
Q	Effective requirement	Domain-specific requirement needed to sustain the declared function/target.
D	Durable capacity	Sustainable ordinary capacity capable of meeting requirement without exceptional bridging.
C_S	Sustainable flexibility	Renewable, planned, ordinary buffering.
C_E	Extraordinary compensation	Exceptional, costly, constrained, temporary, or non-renewable bridging capacity.
F	Observable function	Pre-specified service, reliability, access, or social-reproduction outcome.
G_C	Capacity gap	Q – D, interpreted using domain-specific measurement.

$$G_{C,t} = Q_t - D_t$$

3. The Central Prospective Proposition

SignalRupture predicts that when effective requirement persistently exceeds durable capacity, observable function will initially be preserved to the extent that compensatory capacity can expand. Systems that subsequently close the requirement-capacity gap should reduce extraordinary compensation while maintaining or improving function. Systems that fail to close the gap and whose extraordinary compensation ceases to expand should exhibit subsequent deterioration in pre-specified functional outcomes.

$Q > D \rightarrow \text{compensation response} \rightarrow \text{restoration OR compensation constraint} \rightarrow \text{functional outcome}$

The prediction is conditional, not deterministic. A positive gap does not guarantee deterioration. Compensation may be sustainable; requirement may fall; durable capacity may rise; substitution may emerge; or the original target may change.

4. Prediction P1 — Compensated Continuity

$G_C > 0 \wedge C_E \uparrow \rightarrow F \text{ deteriorates less than under comparable } G_C \text{ with less expandable } C_E$

P1 predicts that compensatory capacity has a temporally protective role. Under comparable requirement-capacity gaps, systems with greater expandable compensation should preserve observable function longer than systems with less compensatory capacity, after relevant domain controls.

This does not imply that compensation is pathological. It predicts only that compensation can bridge a mismatch between requirement and durable capacity.

Primary falsifier:

After credible adjustment for gap magnitude and domain conditions, expandable C_E has no relationship to preservation of F , or systems with less C_E preserve function equally well or better without restoration.

5. Prediction P2 — Surface–Substrate Divergence

$F_A \approx F_B$ does not imply $D_A/Q_A \approx D_B/Q_B$

P2 predicts that two systems can display similar observable function while relying on materially different capacity structures. As compensatory dependence rises, headline output alone should become a weaker indicator of durable sufficiency.

$$\text{Compensatory dependence} = C_E / (D + C_S + C_E)$$

The prediction is strongest when comparable systems produce similar F but one system requires materially greater extraordinary compensation or has a larger unresolved G_C.

Primary falsifier:

Once observable function is controlled, compensation dependence and the Q–D relation contain no reproducible information about structural condition, restoration requirement, or subsequent outcome.

6. Prediction P3 — Compensatory Inversion

$$CI_t = [\Delta C_E, t \leq 0] \wedge [G_C, t > 0] \wedge [\Delta F_t < 0 \text{ or } F_t(t+h) < F_t]$$

P3 is the strongest prospective warning claim. Rising compensation is not the warning. The warning occurs when the structural gap remains unresolved and extraordinary compensation can no longer expand while function subsequently deteriorates.

$$P(F_t(t+h) \downarrow \mid G_C > 0, \Delta C_E \leq 0) > P(F_t(t+h) \downarrow \mid G_C > 0, \Delta C_E > 0)$$

The comparison must be made among systems with sufficiently similar demand, gap severity, and relevant domain conditions. Falling compensation accompanied by restoration is explicitly excluded from CI.

Primary falsifier:

Across repeated cases, plateauing or falling C_E under persistent G_C does not precede greater deterioration than continued expandable compensation, or the association disappears when ordinary domain predictors are included.

7. Prediction P4 — Restoration

$$\Delta D > \Delta Q \text{ for sufficient duration} \rightarrow G_C \downarrow$$

$$G_C \downarrow \rightarrow C_E \text{ dependence} \downarrow \wedge F \text{ stable/improving}$$

P4 freezes the recovery branch. If durable capacity catches requirement, or requirement falls sufficiently, extraordinary compensation should become less necessary without deterioration in function. This prediction prevents SR from interpreting every decline in overtime, borrowing, emergency staffing, or exceptional reserve action as exhaustion.

Primary falsifier:

Measured gap closure repeatedly fails to reduce extraordinary dependence or improve/stabilize function, absent a newly measured requirement increase or other pre-specified explanation.

8. Prediction P5 — Cross-Domain Invariance

P5 predicts that P1–P4 should recur at the level of relational structure across physically heterogeneous domains. The physical measures need not be identical and should not be forced into a universal scalar.

Role	Healthcare	Electricity	Households
Q	Service/workforce requirement	Reliability-conditioned demand	Essential social-reproduction requirement
D	Regular sustainable staffing	Dependable deliverable capacity	Stable recurring resources
C_S	Renewable staffing flexibility	Normal reserve/flexibility	Renewable liquid/risk-sharing buffer
C_E	Extraordinary overtime/purchased staffing	Exceptional emergency substitution	Identified exceptional gap-bridging coping
F	Service volume/access	Reliable delivery	Essential continuity

Primary falsifier:

The directional sequence systematically fails in one or more pre-specified domains, or the grammar can be preserved only by changing the meanings of Q, D, C_E, or F after outcomes are observed.

9. Frozen Relative Temporal Ordering

SR does not impose one universal calendar horizon across hospitals, grids, and households. It freezes relative ordering instead.

t_0 = persistent positive G_C

t_1 = measurable compensation response

t_2 = restoration OR compensation constraint

t_3 = functional outcome

Predicted deterioration branch: $t_0 < t_1 < t_2 \leq t_3$

A domain-specific study may pre-register actual horizons, but it may not reverse this ordering after the outcome. If functional deterioration reliably precedes the supposed compensatory constraint, the CI mechanism is not supported.

10. Healthcare Prospective Translation

Healthcare is the first high-information domain because compensation is directly measurable. The frozen prediction is:

$Q_H > D_H \rightarrow$ extraordinary staffing dependence $\uparrow \rightarrow F_H$ preserved relative to comparable low-buffer systems

Persistent gap + compensation plateau/decline $\rightarrow$ later access/workforce deterioration

Gap closure + compensation decline + access improvement $\rightarrow$ restoration

Candidate F_H outcomes include wait distributions, closures/diversions, backlog, vacancy persistence, absence, turnover, retention, and benchmark attainment. The exact outcomes used in a formal statistical study must be frozen before scoring.

11. Electricity Prospective Translation

Electricity provides the hostile engineered-system test. Normal reserve, storage, routine demand response, and planned flexibility belong to C_S , not C_E . The frozen sequence is:

$Q_E > D_E + C_S, E \rightarrow$ exceptional substitution may preserve F_E

Persistent gap + constrained exceptional substitution → higher subsequent reliability deterioration risk

Durable capacity/flexibility growth $\geq$ requirement growth → extraordinary dependence↓ with F_E maintained

The electricity test fails if SR must relabel ordinary resource-adequacy engineering as extraordinary compensation to obtain confirmation.

12. Household Prospective Translation

The household test requires longitudinal identification. Aggregate debt is not C_E. The frozen sequence is:

Q_HH>D_HH → identified exceptional coping → temporary essential continuity

Persistent gap + coping constraint → later deprivation/arrears/unmet need

Gap closure → reduced exceptional coping + essential function stable/improving

Candidate functional outcomes include food insecurity, persistent arrears, insolvency, housing loss, utility disconnection, unmet essential care, and involuntary crowding.

Purpose and sequence must identify compensation.

13. Prediction Registry

ID	Frozen prediction	Minimum evidence for support	Core falsifier
P1	Expandable compensation preserves function under positive gap.	Temporal gap precedes C_E response; comparable higher-C_E systems preserve F longer.	No protective relation after controls.
P2	Surface function can diverge from substrate sufficiency.	Similar F with materially different gap/compensation states and different later structural outcomes.	Substrate measures add no reproducible information.
P3	Compensatory Inversion precedes deterioration.	Persistent gap + C_E plateau/fall precedes worse F.	No increased deterioration or sequence reversed.

P4	Restoration reduces extraordinary dependence without harming F.	Gap closes; C_E falls; F stable/improves.	Gap closure does not produce predicted recovery relation.
P5	Same relational direction recurs across heterogeneous domains.	At least two heterogeneous domains show the same frozen directional sequence.	Requires post-hoc remapping or systematically fails.

14. Scoring Rules

Score	Definition
Confirmed	Frozen direction, temporal ordering, and mechanism are observed inside the pre-specified test.
Partially Confirmed	Direction appears, but timing, mechanism, breadth, or measurement is incomplete.
Indeterminate	Outcome window incomplete or data inadequate.
Falsified	Countercondition occurs, temporal ordering fails, or the window closes without the predicted relation.

A miss cannot be rescued by moving the observation window, changing the outcome, redefining compensation, or selecting a favorable subgroup after the result is known.

15. Predictive Validation Levels

Level	Status requirement
Level 0 — Conceptual Grammar	Coherent relational architecture.
Level 1 — Operational Invariance	Stable mapping across heterogeneous domains.
Level 2 — Domain Predictive Validity	Prospective directional prediction succeeds in at least one domain.
Level 3 — Cross-Domain Predictive Validity	Same frozen grammar predicts prospectively in at least two heterogeneous domains.
Level 4 — Predictive Metatheoretical Validation	Cross-domain prospective prediction plus restoration prediction, incremental value, robustness, and independent replication.

As of August 2026, the Q–D–C–F program is classified at Level 1 with partial descriptive evidence toward Level 2. This manuscript freezes the tests required for movement beyond Level 1.

16. Incremental Prediction Requirement

Correct prediction alone is not sufficient for the strongest validation claim because established domain sciences may already predict the same outcomes. Formal studies should compare a credible baseline model with an SR-augmented model.

$$M_{0,d}: F_{-}(i,t+h) = f_{-}d(\text{best established domain predictors})$$

$$M_{SR,d}: M_{0,d} + G_C + C_E + \Delta C_E + CI + \text{restoration state}$$

$$\Delta \text{Perf}_d = \text{Perf}(M_{SR,d}) - \text{Perf}(M_{0,d})$$

Level 4 requires reproducible positive incremental value in more than one heterogeneous domain. Appropriate performance metrics depend on the outcome and may include forecast error, calibration, discrimination, lead time, or decision utility.

17. Lead-Time Prediction

A diagnostic metatheory becomes especially valuable if it identifies deterioration earlier than conventional output indicators. The protocol therefore freezes a secondary prediction:

$$\text{LeadTime}_{SR} > \text{LeadTime}_{\text{output-only}}$$

If substrate and compensation measures consistently identify risk before headline function deteriorates, SR gains diagnostic-predictive value even when ultimate outcome classification is also achievable by conventional methods closer to failure.

18. Restoration as a Mandatory Counterprediction

The prospective program is invalid if it scores only deterioration. Every formal test must identify restoration cases where possible. A system that closes G_C , rebuilds durable slack, reduces C_E , and maintains or improves F is a successful prediction of the restoration branch.

Restoration is evidence for the grammar, not an escape from falsification. Conversely, if SR repeatedly predicts deterioration where restoration occurs under its own measured variables, the model must lose score.

19. Negative Controls

- High compensation with no positive structural gap and no subsequent deterioration.
- Falling compensation caused by durable restoration and improving function.
- High electricity reserve use that is normal and renewable under planning criteria.
- Household borrowing paid routinely without an essential-resource gap.
- Hospital overtime that remains stable and renewable without worsening retention, absence, vacancies, cost, or access.
- Systems with similar G_C but radically different outcomes explained fully by known domain variables, with no residual C_E effect.

20. Prohibition on Universal Scalarization

The prospective grammar does not require hospital hours, megawatts, household dollars, and access outcomes to be converted into a single SR score. Such a conversion would reintroduce unsupported pseudo-precision. Cross-domain validation occurs through relational direction, temporal ordering, and standardized predictive performance.

Cross-domain invariance $\neq$ cross-domain unit equivalence

21. Falsification Architecture

- P1 fails if compensation does not prospectively buffer function under comparable gaps.
- P2 fails if substrate state contains no reproducible information beyond observed function.
- P3 fails if compensation constraint under unresolved gaps does not precede increased deterioration.
- P4 fails if genuine gap closure does not reduce extraordinary dependence while preserving function.

- P5 fails if the same roles cannot be retained across domains without post-hoc semantic changes.
- The stronger metatheory claim fails if SR adds no incremental predictive or decision value beyond mature domain models.
- Any forecast rescued by changing dates, thresholds, outcomes, or groups after observation is scored as a miss.

22. What Would Constitute Strong Evidence

- A pre-registered healthcare panel shows P1 and P3 prospectively and predicts access deterioration earlier than output-only baselines.
- A pre-registered electricity panel independently reproduces the same relational direction using non-human mechanisms.
- At least one domain shows the restoration branch exactly as frozen.
- SR-augmented models improve out-of-sample performance beyond strong domain baselines.
- Results survive reasonable alternative Q definitions and sensitivity analysis.
- Independent researchers reproduce the mapping and predictive gain.

23. What Would Constitute Failure

The grammar should be narrowed or rejected as a predictive metatheory if prospective tests show that the compensation state is epiphenomenal, if the temporal ordering fails, if restoration does not behave as predicted, or if established domain models absorb all apparent SR value. Operational recurrence alone would then remain a synthesis or diagnostic vocabulary rather than evidence of a predictive metatheory.

24. Relationship to the SR Collapse Warning

This prediction is separate from the SR Collapse Warning. The Collapse Warning freezes a particular prospective sequence and observation windows for Canada and the United States. The Structural Grammar Prediction freezes a more general relational claim that

can be tested repeatedly in multiple systems and at multiple scales. The two should not be rewritten to rescue one another.

Collapse Warning = future case forecast

Structural Grammar Prediction = general relational forecast

If both succeed independently, their evidentiary value compounds. If one fails, the other must be scored on its own terms.

25. Canonical Prediction Statement — Frozen August 2026

SignalRupture predicts that when effective system requirement persistently exceeds durable capacity, observable function will initially be preserved to the extent that compensatory capacity can expand. During this phase, surface performance will be an incomplete indicator of durable structural sufficiency. Systems that subsequently close the requirement-capacity gap should reduce extraordinary compensation while maintaining or improving function. Systems that fail to close the gap and whose extraordinary compensation ceases to expand should exhibit subsequent deterioration in pre-specified functional outcomes. This directional structure should recur across heterogeneous domains even though the physical forms of requirement, durable capacity, compensation, restoration, and function differ. The prediction is materially weakened or falsified if compensation state has no prospective relationship to function under persistent gaps, if restoration does not reverse extraordinary dependence as predicted, or if the cross-domain relation can be preserved only through post-hoc redefinition.

26. Frozen Status at Issuance

Claim	August 2026 status
Structural grammar is conceptually specified	Established within SR.
Three-domain operational translation	Achieved.
Direct descriptive compensated-continuity evidence	Partial; strongest in Canadian healthcare.
Prospective Compensatory Inversion	Unvalidated.

Prospective restoration prediction	Unvalidated.
Cross-domain prospective prediction	Unvalidated.
Incremental predictive value	Unvalidated.
Predictive metatheoretical validation	Not yet achieved.

27. Update Protocol

Every future scoring manuscript must reproduce the Canonical Prediction Statement and Prediction Registry unchanged. It must then append: new observations; measurement revisions; pre-specified outcome scores; alternative explanations; temporal-order assessment; comparison with domain baselines; restoration cases; falsifications; and any newly formulated prediction stated separately from the August 2026 version. The original prediction may not be silently rewritten.

28. Conclusion

The SignalRupture Structural Grammar now moves from operational recurrence to prospective risk. The prediction is intentionally simple enough to lose. Systems with unresolved requirement-capacity gaps should initially preserve function where compensation can expand. Surface function should therefore sometimes overstate durable sufficiency. If durable restoration closes the gap, extraordinary dependence should fall without functional deterioration. If the gap persists and compensation becomes constrained, deterioration should follow. The same directional architecture should recur across heterogeneous systems if SR is genuinely structural rather than domain-specific.

The future evidence may support, narrow, or falsify this claim. That is the purpose of freezing it. SR does not gain predictive metatheoretical validation because the grammar is plausible, because three domains can be mapped onto it, or because future deterioration occurs somewhere. It gains that status only if the pre-specified relations occur prospectively, in the correct temporal order, across heterogeneous domains, while restoration behaves as predicted and the SR representation adds value beyond established domain science.

The August 2026 commitment is therefore explicit: the grammar will not be judged by whether SR can explain the future after it arrives. It will be judged by whether the future behaves as SR said it should before the outcome was known.

Appendix A. Frozen Prediction Registry

ID	Prediction	Status at issuance
P1	Positive persistent gap plus expandable compensation preserves function relative to comparable low-compensation systems.	Prospective
P2	Similar surface function can coexist with materially different substrate sufficiency and compensation dependence.	Prospective with descriptive support
P3	Persistent gap plus plateauing/falling extraordinary compensation predicts subsequent functional deterioration.	Prospective
P4	Gap closure predicts falling extraordinary compensation with stable/improving function.	Prospective
P5	P1–P4 recur relationally across heterogeneous domains without post-hoc remapping.	Prospective

Appendix B. Minimum Future Scorecard

Domain	P1	P2	P3	P4	Incremental prediction
Healthcare	Pending	Partial baseline support	Pending	Pending	Pending
Electricity	Pending	Operational support	Pending	Pending	Pending
Households	Pending	Operational/distributional support	Pending	Pending	Pending

